

鄭聖文 Sheng-Wen (Colin) Cheng

 GitHub |  LinkedIn |  Website |  ORCID |  Google Scholar |  shengwen1997.tw@gmail.com

Employment

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| NVIDIA
System Software Engineer | Feb. 2024 - Present
Taipei, Taiwan |
| <ul style="list-style-type: none">• Develop Platform Security Controller (PSC) firmware to protect Linux for Tegra and autonomous-vehicle products• Implement security features including boot-image measurement, authentication, encryption, decryption, etc. | |
| Avilon Intelligence
Embedded Systems Engineer | Sept. 2018 - Mar. 2021
Tainan, Taiwan |
| <ul style="list-style-type: none">• Designed an ARM Cortex-A72-based onboard computer for UAVs, covering PCB layout and system bring-up• Developed a 4G LTE system-on-module (SoM) baseboard for UAV remote control and video streaming• Built vision-based tag-detection and localization software to enable autonomous drone landing | |

Education

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| The University of Texas at Austin
M.S. in Computer Science | Aug. 2025 - Dec. 2026
Texas, USA |
| <ul style="list-style-type: none">• Focus on machine learning, optimization, and scientific computing for intelligent autonomous systems.• Publication: American Control Conference (ACC 2026)• Presentation: Open Source Summit North America 2026 (Linux Foundation Event) | |
| National Taiwan University
Ph.D. Student in Electrical Engineering (Withdrew) | Sept. 2022 - May 2024
Taipei, Taiwan |
| <ul style="list-style-type: none">• Transitioned to industry to gain professional experience while remaining involved in academic activities• Presentation: Embedded Open Source Summit 2024 (Linux Foundation Event) | |
| National Yang Ming Chiao Tung University
M.S. in Robotics | Sept. 2019 - Nov. 2021
Hsinchu, Taiwan |
| <ul style="list-style-type: none">• Master Thesis: Design of Indoor-Outdoor Smooth Transferable Unmanned Aerial Vehicle• Primary contributor to a collaborative research project with the Taiwan Space Agency (TASA)• Two conference papers on quadrotor H_∞ control and error-state Kalman filter for localization after graduation | |
| Providence University
B.E. in Computer Science and Information Engineering | Sept. 2015 - June 2019
Taichung, Taiwan |
| <ul style="list-style-type: none">• First Prize in the graduation project with optical-flow velocity estimation and altitude control for quadrotor.• Conducted real-time UGV research with a paper published at the Science and Information Conference (SAI), UK. | |

Academic Service

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- Invited Reviewer for the **American Control Conference (ACC 2026)**

Certificate Coursework

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| MITx MicroMasters Program in Statistics and Data Science
Certificate Program by Massachusetts Institute of Technology [Program Letter] | Jan. 2024 - Dec. 2025
MITx |
| <ul style="list-style-type: none">• Coursework: Probability, Statistics, Machine Learning, Time Series Analysis, and proctored Capstone Exam. | |

Projects

NCRL Flight Control [\[Video\]](#)

- Leading developer of the holistic system to build an agile quadrotor with nonlinear control and state estimation
- Project licensed to the Taiwan Space Agency (TASA) for research on aggressive quadrotor maneuver control

Tenok: A Linux-like RTOS for Robotics and IoT [\[GitHub\]](#)

- Built a POSIX-compliant RTOS optimized for ARM Cortex-M, adopting Linux-inspired kernel designs for robotics
- Implemented Unix-style OS features including pthreads, mutexes, semaphores, pipes, message queues, and signals
- Added Linux-like kernel mechanisms such as kthreads, wait queues, SLAB, kfifo, tasklets, printk, etc.

PU-01: Twin-boom Fixed-wing UAV [\[Link\]](#)

- Determined aerodynamic parameters, performed airfoil selection, and conducted stability analysis using XFLR5
- Created 3D CAD models of the structural components and fabricated the complete airframe

Semu: Minimalist RISC-V System Emulator [\[GitHub\]](#)

- Contributed to VirtIO hardware virtualization of GPU, input devices, storage, and random number generator

Publications

- [1] **S.-W. Cheng** and T.-H. Cheng, “Data-Driven Estimation of Quadrotor Motor Efficiency via Residual Minimization,” *American Control Conference (ACC)*, New Orleans, USA, 2026 (to appear). [arXiv:2510.11388](#)
- [2] **S.-W. Cheng** and Y.-H. Huang, “A Computationally Efficient GNSS/INS Design of Multirotor based on Error-state Kalman Filter,” *62nd Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*, Tsu, Japan, 2023. [\[Link\]](#)
- [3] **S.-W. Cheng** and H.-A. Hung, “Robust State-Feedback H_∞ Control of Quadrotor,” *International Automatic Control Conference (CACS)*, Kaohsiung, Taiwan, 2022. [\[Link\]](#)
- [4] S.-W. Wang, **S.-W. Cheng**, and C.-C. Huang, “Puyuma: Linux-based RTOS Experimental Platform for Constructing Self-Driving Miniature Vehicles,” *Science and Information Conference (SAI)*, London, United Kingdom, 2018. [\[Link\]](#)

Presentations

- [1] “Demystifying VirtIO-GPU: Building a Graphics Virtualization Bridge from Scratch,” Open Source Summit North America (OSSNA 2026, **Linux Foundation Event**), Minneapolis, USA, 2026. [\[Link\]](#) [\[PDF\]](#)
- [2] “Building a Quadrotor Simulator with Python - Modeling, Simulation, and Control,” Open Tech Conference (OpenTechConf 2025), Hong Kong, 2025. [\[Link\]](#) [\[PDF\]](#)
- [3] “Crafting a Vision-Aided Software Stack for UAV,” Embedded Open Source Summit (EOSS 2024, **Linux Foundation Event**), Seattle, USA, 2024. [\[Link\]](#) [\[PDF\]](#)
- [4] “Creating a Linux-like Real-Time Operating System for Quadrotor Drones,” Conference for Open Source Coders, Users, and Promoters (COSCUP 2024), Taiwan, 2024. [\[Link\]](#) [\[PDF\]](#)
- [5] “Tenok: Build a real-time operating system for Robotics,” Conference for Open Source Coders, Users, and Promoters (COSCUP 2023), Taiwan, 2023. [\[Link\]](#) [\[PDF\]](#)

Invited Talks

- [1] “Trends in Machine Learning for Unmanned Aerial Vehicle Applications,” Mobile Open Platform (MOPCON 2024), **Keynote speaker**, Taiwan, 2024. [\[Link\]](#) [\[PDF\]](#)
- [2] PEGATRON Corporation: “Trends and lessons learned in deep learning and generative AI applications for UAV,” Taipei, Taiwan, 2024.